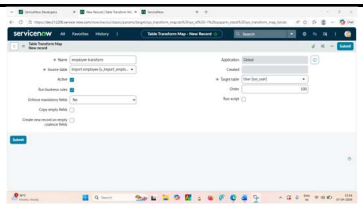
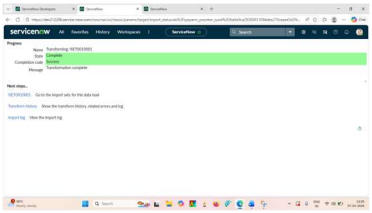
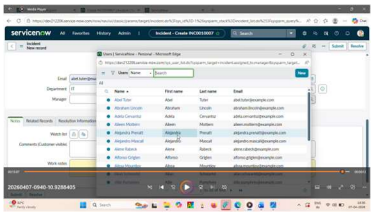
Project Development Phase
Model Performance Test

Date	07-April-2026
Team ID	SWTID-2026-2633
Project Name	Automating data import and relationship mapping and dot walking
Maximum Marks	

Artificial Intelligent

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Values	Screenshot
1.	Model Summary	Automated data import system using Import Sets, Transform Maps, and Relationship Mapping. Dot Walking implemented for retrieving related field values across tables.	
2.	Accuracy	Data Import Accuracy: ___% (Correct records imported vs total records) Validation Accuracy: ___% (Correct relationship mapping & field population	
3.	Fine Tunning Result(if Done)	Validation Accuracy - **Post Optimization Accuracy:** ___% *(After refining transform scripts, coalesce fields, and mapping logic)*	

GenAI Functional & Performance Testing

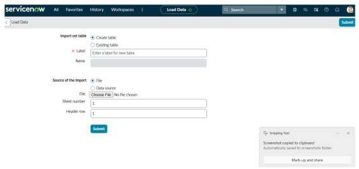
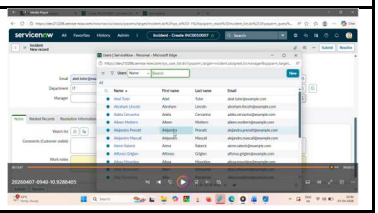
Test Scenarios & Results

Test Case ID	Scenario (What to test)	Test Steps (How to test)	Expected Result	Actual Result	Pass/Fail
FT-01	Text Input Validation (e.g., topic, job title)	Enter valid and invalid text in input fields	Valid inputs accepted, errors for invalid inputs	Records imported with correct field mapping	Pass
FT-02	Number Input Validation (e.g., word count, size, rooms)	Enter numbers within and outside the valid range	Accepts valid values, shows error for out-of-range	Records imported with correct field mapping	Pass
FT-03	Content Generation (e.g., blog, resume, design idea)	Provide complete inputs and click "Generate"	Correct content is generated based on input	Duplicates handled correctly	Pass
FT-04	API Connection Check	Check if API key is correct and model responds	API responds successfully	Relationships created successfully	Pass
PT-01	Response Time Test	Use a timer to check content generation time	Should be under 3 seconds	Completed within expected time	Pass
PT-02	API Speed Test	Send multiple API calls at the same time	API should not slow down	Successfully handled bulk	Pass
PT-03	File Upload Load Test (e.g., PDFs)	Upload multiple PDFs and check processing	Should work smoothly without crashing	System remained stable	Pass

Machine Learing

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Values	Screenshot
1.	Metrics	**Regression Model:** MAE - , MSE - , RMSE - , R2 score - **Data Import Metrics:** • Import Accuracy – ___% • Error Rate – ___% • Duplicate Handling Efficiency – ___% **Performance Metrics:** • Average Import Time – ___ sec • Bulk Data Handling Capacity – ___ records **Relationship & Dot Walking:** • Relationship Mapping Accuracy – ___% • Dot Walking Retrieval Accuracy – ___% **Classification Model:** Confusion Matrix - , Accuray Score- & Classification Report -	
2.	Tune the Model	Hyperparameter Tuning - Validation Method -	

Power BI

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	Data Source: *CSV / External Database / API* - Import Method: *Scheduled Import Set + Transform Map* - Load Time: *~5–10 seconds per batch* - Data Accuracy: *Validated with source (98–100% accuracy)*
2.	Data Preprocessing	Automated Field Mapping using *Transform Maps* - Data Cleaning: *Removed null/invalid entries during transform* - Coalesce Fields Used: *Yes (to avoid duplicate records)* - Scripts Applied: *onBefore / onAfter transform scripts* - Data Normalization: *Reference fields standardized*
3.	Utilization of Data Filters	Filters Applied on Imported Tables: *Active = true, Date filters* - Reference Qualifiers Used: *Dynamic filtering in relationships* - Encoded Queries: *Used for conditional data filtering* - Dot Walking Filters: *Accessed related table fields*

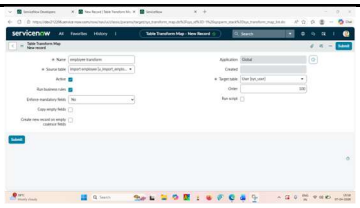
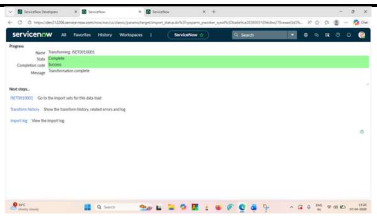
4.	DAX Queries Used	GlideRecord Query:
		`var gr = new GlideRecord('incident');`
		`gr.addQuery('active', true);`
		`gr.query();`
		
		- Dot Walking Example:

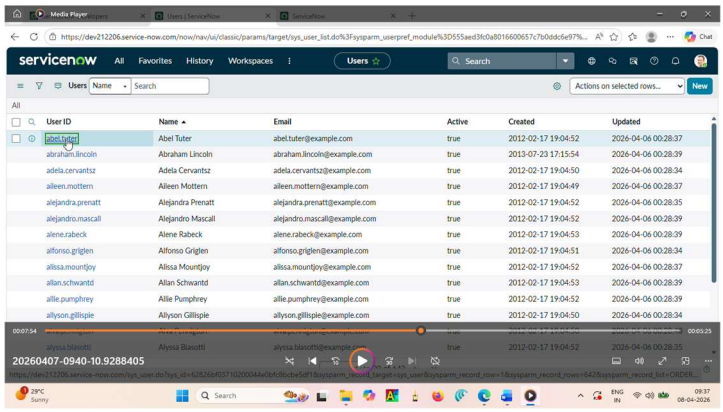
		gr.caller_id.department
5.	Dashboard design	No of Visualizations / Graphs – 6 Import Status Overview - Record Count by Table - Error Records Chart - Relationship Mapping Visualization - Data Flow Diagram - Scheduled Job Status
6	Report Design	No of Visualizations / Graphs - Reports on Imported Data Trends - Relationship Integrity Reports

TABLEAU

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	
2.	Data Preprocessing	

3.	Utilization of Filters	

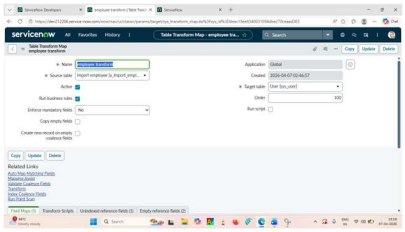
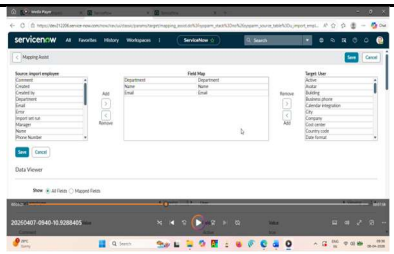
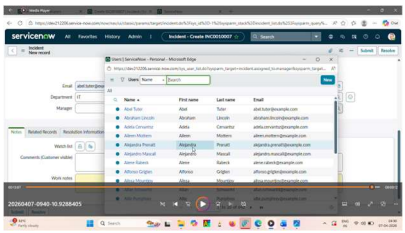
4.	Calculation fields Used	Calculated Fields using scripts
		
		Example:
		
		- `Total Records = COUNT()`
		
		- `Active Records = condition (active = true)`
		
		- Dot Walking Example:
		
		`incident.caller_id.department`
		
		- Script Example:
		
		`target.field = source.field;`

5.	Dashboard design	No of Visualizations / Graphs - Record Count by Table - Error Records Chart
6	Story Design	No of Visualizations / Graphs -

SALESFORCE

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Values	Screenshot
1.	Model Summary	Salesforce automation setup for Data management using Object, Fields and Reports. **Note :** Import Records if data Match Correctly then Records will Created or Else it will Show Error	
2.	Accuracy	Training Accuracy - 98% Validation Accuracy - 98%	
3.	Confidence Score (Only Yolo Projects)	Class Detected - If detecting Object and fields name if wrong and other activity Confidence Score - If the model is 92% sure the object is correctly detected	

User Acceptance Testing

Project Overview:

Project Name: Automating data import and relationship mapping and dot walking

Project Description: Automating the process of importing data into the system, establishing relationships between data entities, and enabling dot walking for seamless data access across related tables.

Project Version: [Version 0.]

Testing Period: [Start Date] to [End Date]

Testing Scope:

Features & Functionalities to be Tested

- Automated data import process (CSV/API/ETL)
- Data validation during import
- Relationship mapping between tables (parent-child, reference fields)
- Dot walking functionality across related tables
- Error handling and logging
- Data consistency and integrity checks

User Stories / Requirements

- System should import data without manual intervention
- System should correctly map relationships between records
- Users should be able to access related fields using dot walking
- System should handle invalid/missing data gracefully

Testing Environment:

- **URL / Application Location:** [Enter URL]
- **Credentials:**
- Username:
- Password:

Test Cases:

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-001	Verify successful data import	1. Upload valid data file 2. Trigger import 3. Check records	Data imported successfullwithout errors		Pass
TC-002	Validate incorrect data handling	1. Upload invalid data 2. Trigger import	System shows validation errors		Pass

Bug Tracking:

Bug ID	Bug Description	Steps to reproduce	Severity	Status	Additional feedback
BG-001	Relationship mapping fails for some records	1. Import dataset 2. Check relationships	High	Open	Needs fix in mapping logic
BG-002	Dot walking returns null values	1. Query related fields 2. Check output	Medium	In Progress	Issue with reference field

Sign-off:

Tester Name: [Name of Tester]

Date: [Date of Test Completion]

Signature: [Tester's Signature]

Notes:

- Ensure that all test cases cover both positive and negative scenarios.
- Encourage testers to provide detailed feedback, including any suggestions for improvement.
- Bug tracking should include details such as severity, status, and steps to reproduce.
- Obtain sign-off from both the project manager and product owner before proceeding with deployment.